AI-Based Electrical Fire Prevention System Using IoT Sensors and Machine Learning

Surabhi Kerketta1, G.Rithika2, Vagitha Rifana3,Dr. G. Revathy4

Department of Computer Science and Engineering(Artificial Intelligence and Machine Learning), Vels Institute of Science,Technology and Advanced Studies,

Pallavaram, India

Abstract:

Electrical fires are often caused by electrical and environmental conditions and are not detected until the smoke and flames of the fire start. To address this issue, we have created a prototype that monitors the temperature, voltage, current and smoke of an electrical device using the ESP32 IoT platform. An machine learning model classifies the risk of the situation into four levels of risk safe,warning,high risk or critical.We created a data set of 6,400 samples to train the machine learning model. The random forest model that was trained with the data set achieved an accuracy of 99.06%. However, for the embedded device that we created, a decision tree model was used which achieved an accuracy of 98.12%. When the system detects a state that indicates a potential fire, it will trigger a relay switch that cuts the power to the electrical device. In addition, a buzzer will sound and an SMS message will be sent to a designated phone number via the SIM800L GSM module. All sensor readings and events that occur are also logged. We tested our prototype in a controlled environment. This is a student prototype and is not intended to be used for high current applications.

Keywords: Fire prevention, ESP32, Internet of Things, Machine learning, Decision Tree, Random Forest, Multi-sensor monitoring GSM alert

Institute of Computer Science and Engineering, SIMATS Engineering, Chennai.
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